Tom Marty

3rdcore.github.io tom.marty@mila.quebec Google Scholar

Education

Ph.D. in Machine Learning — Jan. 2024 – Mar. 2026
Mila, Université de Montréal - GPA 4.00 (leave of absence) — *Montréal, Canada*

M.Sc. in Machine Learning — Sep. 2021 – Aug. 2023
Polytechnique Montréal - GPA 3.91 — *Montréal, Canada*

B.Sc. in Computer Science, minor in Applied Mathematics — Sep. 2018 – Aug. 2021
***X 2018**, Ecole Polytechnique - GPA 3.84* — *Palaiseau, France*

Advanced Preparatory Class for Competitive Exams — Sep. 2016 – Aug. 2018
Lycée Jean-Baptiste Say - GPA 4.00 - Top 0.1% national — *Paris, France*

Research Interest

- **Broad interest** : Deep Learning, Generative modeling and Operational Research
- **Methodological interest** : Flow-based model, Causality, Generalization, Information Theory
- **Applications** : ML for biology, Robust Machine Learning, Open-Ended Decision Making

Industry and Academic Experience

Visiting Researcher — Apr. 2023 – Sept. 2023
ServiceNow Research — *Montréal, Canada*
- Developped WorkArena (ICML2024): an open-source Benchmark and Gym environment for evaluating Agent at solving common-knowledge tasks on a Web Browser

Research Supervisor — Jan. 2022 – Sept. 2022
Corail Research Group — *Montréal, Canada*
- Supervised five interns on the development of the open-source project SeaPearl
- Teaching Assistant for the course INF8215 given by Quentin Cappart in Fall 21 and Fall 22

Research Engineer Intern — Jan. 2021 – Sept. 2021
Corail Research Group — *Montréal, Canada*
- Developped *SeaPearl* : an open-source RL-driven generic Constraint Programming solver
- Used Deep Q-networks and Heterogeneous GNNs to approximate optimal decision process

Software Engineer Intern — Jun. 2020 – Sept. 2020
Dronisos, drone light show company — *Bordeaux, France*
- Developed *Harmony*, a Physics based meta-heuristic that secures massive drone swarms
- *Harmony* - currently in use - reduced the allocated securing time from 2 weeks (handmade) to 2 seconds
- Achieved automatic securing on the company first 1000 drones choreography (+500k$ show)

Conference and Journal Publications

A Compression Perspective on Simplicity Bias
Tom Marty*, Eric Elmoznino, Tejas Kasetty, Léo Gagnon, Mizu Nishikawa-Toomey, Sarthak Mittal, Guillaume Lajoie, Dhanya Sridhar
*under review

In-Context Learning and Occam's Razor
Eric Elmoznino*, **Tom Marty***,Tejas Kasetty, Leo Gagnon, Sarthak Mittal, Mahan Fathi, Dhanya Sridhar, Guillaume Lajoie
International Conference on Machine Learning (ICML 2025) Code PDF

Does Learning the Right Latent Variables Necessarily Improve In-Context Learning?
Sarthak Mittal, Eric Elmoznino, Leo Gagnon, Sangnie Bhardwaj, **Tom Marty**, Dhanya Sridhar, Guillaume Lajoie
International Conference on Machine Learning (ICML 2025) PDF

Next-Token Prediction Should be Ambiguity-Sensitive: A Meta-Learning Perspective
Leo Gagnon, Eric Elmoznino, Sarthak Mittal, **Tom Marty**, Tejas Kasetty, Dhanya Sridhar, Guillaume Lajoie
International Conference on Machine Learning (FoMo@ICML 2025) PDF

The BrowserGym Ecosystem for Web Agent Research
Thibault Le Sellier De Chezelles, Maxime Gasse, Alexandre Drouin, Massimo Caccia, Léo Boisvert, Megh Thakkar, **Tom Marty**, Rim Assouel, Sahar Omidi Shayegan, Lawrence Keunho Jang, Xing Han Lù, Ori Yoran, Dehan Kong, Frank F. Xu, Siva Reddy, Quentin Cappart, Graham Neubig, Ruslan Salakhutdinov, Nicolas Chapados, Alexandre Lacoste
Transactions on Machine Learning Research (TMLR 2025) . Code PDF

Learning and Fine-Tuning a Generic Value-Selection Heuristic Inside a Constraint Programming Solver
Tom Marty*, Léo Bois-Vert*, Tristan François, Pierre Tessier, Louis Gautier, Léo-Boisvert, Louis-Martin Rousseau, Quentin Cappart, Constraint **Journal** 2024. Code PDF

WorkArena: How Capable Are Web Agents at Solving Common Knowledge Work Tasks?
Alexandre Drouin, Maxime Gasse, Massimo Caccia, Issam H Laradji, Manuel Del Verne, **Tom Marty**, Léo Boisvert, Megh Thakkar, Quentin Cappart, David Vazquez, Nicolas Chapados, Alexandre Lacoste
International Conference on Machine Learning (ICML 2024) . Code PDF

The Unsolved Challenges of LLMs as Generalist Web Agents: A Case Study
Rim Assouel*, **Tom Marty***, Massimo Caccia, Issam H. Laradji, Alexandre Drouin, Sai Rajeswar, Hector Palacios, Quentin Cappart, David Vazquez, Nicolas Chapados, Maxime Gasse, Alexandre Lacoste
Foundation Models for Decision Making Workshop (FMDM@NeurIPS 2023) . Code PDF

Learning a Generic Value-Selection Heuristic Inside a Constraint Programming Solver
Tom Marty*, Tristan François, Pierre Tessier, Louis Gautier, Louis-Martin Rousseau, Quentin Cappart
Distinguished paper, Constraint Programming (CP 2023). Code PDF

You can also find the most up-to-date publications on my Google Scholar page.

Other Projects

Discrete flow matching for protein co-design aimed at pMHC:TCR binding Sep. 2025 –

Dissection of concept acquisition in Protein Language Model Sep. 2025 –

BrowserGym : an Open-Source Benchmark for evaluating Web Agents | *Python* Apr. 2023 – Mar. 2024

- Paper accepted at ICML 2024 Vienna, presented at NVIDIA GTC 2024

SeaPearl : an Open-Source RL-driven Constraint-Programming Solver | *Julia* Fev. 2021 – Jul. 2023

- Paper accepted at CP2023, Toronto

Adversarial Attacks on Sentiment Classification models | *Python, HuggingFace* Fev. 2022 – May. 2022

- Adversarial fine-tunning on large NLP models : Eleuther AI GPT 125M/1.3B/2.7B/6B parameters
- Evidence of a correlation between scaling and robustness against increasingly subtle Adversarial Attacks

Diffusion Geodesic distance for non-linear dimensionality reduction | *Python* Oct. 2021 – Jan. 2022

- Approximated the geodesic distance using a diffusion process over the manifold
- Proposed a new data visualization algorithm based on Multi-Dimentionnal Scaling and Diffusion Geodesic

Autonomous Drone Swarm Deployment - DGA contest | *Python, PyTorch* Nov. 2020 – Mar. 2021

- Multi-agent Q-Learning method for deployment optimization
- Density-Based Spatial Clustering for point of interest detection

Realtime 3D Deep Motion Capture | *C++, OpenCV, PyTorch* Oct. 2020 – Dec. 2020

- Implemented a method of inferring a full character's 3d pose using only a camera as an input
- Inspired by a EECV 2020 research paper to implement the algorithm

Sketch-based Shape Retrieval | *Python, C++, OpenGL* Sep. 2020 – Dec. 2020

- Implemented a method to find any specific 3d model in a database using a drawing as an input
- Succeeded to faithfully retrieve several simple 3D shapes by using a single drawing given by a user

Visit 🔗 my website to know more about me and my projects!

Honors and Awards

FRQNT doctoral training scholarship: 25000$ (x4 years)	Mar. 2025
Distinguished Paper Award at CP conference	Sept. 2023
MITACS Accelerate scholarship: 30000$	Mar. 2023
Oustanding Investment Mention, Ecole Polytechnique de Paris	Jul. 2022
Vallet Fondation scholarships for outstanding CPGE students: 5000€ (x2 years)	2018

Teaching Experience

Teaching Assistant Fall 2025
IFT6390: Fundamentals of machine learning

Teaching Assistant Fall 2022
INF8215, Artificial Intelligence : Algorithms and methods

Teaching Assistant Fall 2021
INF8215, Artificial Intelligence : Algorithms and methods

Teaching Assistant Nov. 2018 – Mar. 2019
Ministry of National Education *France*

- Responsible for a group of up to 20 undergraduate students during scientific workshops
- Worked alongside the academic team to prepare students for entrance exams

Community Involvement and Service

Tea Talks Committee Member – Research Seminar held at Mila 2025

Reviewer for NeurIPS 2025 NewInML Workshop 2025

Reviewer for NeurIPS 2024 CALM Workshop, MAIS 2024, HRAIM 2024 2024

Reviewer for CP2023 2023

President of Nuit du Styx festival 2020

- Oversaw general organization and logistics for an electronic music festival with over 2000 attendees

Member of Rethorix, Public Speaking Club 2019

- Organized an eloquence contest between schools in the Plateau de Saclay area

Skills & Hobbies

Languages: French : Native | English : Fluent | Russian : Primary
Developer Toolbox: Pytorch, Lightning, Git, Hydra, HPC, WandB, SLURM, CI Testing
Programming Languages: Python, Julia, C++, R

Activities: Outdoor climbing, Surfing, Snowboarding, mountain hiking, UAV robotics

Reference

Prof. Dhanya Sridhar (Ph.D. advisor)
Assistant Professor at UdeM, Core academic member at MILA - AI CIFAR Chair holder
Email : dhanya.sridhar@mila.quebec

Dr. Alexandre Lacoste
Staff Research Engineer, ServiceNow Research
Email : alexandre.lacoste@servicenow.com

Prof. Quentin Cappart (M.Sc. advisor)
Assistant Professor at Polytechnique Montréal
Email : quentin.cappart@polymtl.ca